

Aqueous extract of *Trigonella foenum-graecum* L. ameliorates additive urotoxicity of buthionine sulfoximine and cyclophosphamide in mice

K. Bhatia, M. Kaur, F. Atif, M. Ali, H. Rehman, S. Rahman, S. Raisuddin *

Department of Medical Elementology and Toxicology, Jamia Hamdard (Hamdard University), New Delhi 110 062, India

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Abstract

Cyclophosphamide (CP) is a commonly used anti-cancer drug which causes toxicity by its reactive metabolites such as acrolein and phosphoramidate mustard. In the present study modulation of toxicity caused by concomitant exposure to CP and L-buthionine-SR-sulfoximine (BSO) by fenugreek (*Trigonella foenum-graecum* L.) extract was evaluated by measuring lipid peroxidation (LPO) and anti-oxidants in urinary bladder in mice. Fenugreek, a common dietary and medicinal herb, showed protective effect not only on LPO but also on the enzymatic anti-oxidants. CP-treated animals exhibited a significant decrease in the activities of glutathione S-transferase (GST), glutathione reductase (GR), glutathione peroxidase (GP) and catalase (CAT) when compared to the controls. Level of reduced glutathione (GSH) was also reduced with an increase in LPO in CP-treated animals. BSO treatment depicted an additive toxic effect in CP-treated animals. Pre-treatment of herbal extract restored activities of all the enzymes and thus showed an overall protective effect on additive effect of CP and BSO. Restoration of GSH by extract treatment may play an important role in reversing CP-induced apoptosis and free radical-mediated LPO in urinary bladder. Fenugreek, known for its hypoglycemic, anti-inflammatory and immunomodulatory activity, may be a promising protective medicinal herb for consideration in complementary therapy in cancer patients under chemotherapeutic interventions. © 2006 Elsevier Ltd. All rights reserved.

Keywords: Cyclophosphamide; BSO; Urinary bladder anti-oxidants; Fenugreek extract; Protection

1. Introduction

Cyclophosphamide (CP) is an alkylating agent widely used in anti-neoplastic therapy (Baumann and Preiss, 2001; Fleming, 1997). It is effective against a variety of can-

cers such as lymphoma, myeloma and chronic lymphocytic leukemia (Baumann and Preiss, 2001). However, CP intake is associated with alopecia, cardiac damage, gonadotropy, hemorrhagic cystitis, hematopoietic depression, nausea, vomiting, and carcinogenicity (Fleming, 1997; Mirkes, 1985). It is also reported to cause lung toxicity (Patel and Block, 1985). The urological side effects of CP are major limiting factor in its use. These effects include transient irritative voiding symptoms such as dyspsia, hemorrhagic cystitis, bladder fibrosis, necrosis, contracture and vesicometral flux (Levine and Richie, 1989). CP treatment also results in the production of reactive oxygen species (ROS), which cause peroxidative damage to urinary bladder and other vital organs (Patel, 1987; Sulkowska et al., 1998). Role of tissue anti-oxidants becomes important in the prevention of such damage. A number of natural products and synthetic compounds have been shown to reduce CP toxic-

Abbreviations: ANOVA, analysis of variance; BSO, L-buthionine-SR-sulfoximine; CAT, catalase; CDNB, 1-chloro-2,4-dinitrobenzene; CP, cyclophosphamide; DTNB, dithio-bis-2-nitrobenzoic acid; EDTA, ethylenediamine tetra acetic acid disodium salt; γ -GCS, γ -glutamylcysteine synthetase; GP, glutathione peroxidase; GR, glutathione reductase; GSH, reduced glutathione; GST, glutathione S-transferase; NADPH, nicotinamide adenine dinucleotide phosphate reduced; LPO, lipid peroxidation; PE, plant extract; PMS, post-mitochondrial supernatant; ROS, reactive oxygen species; SE, standard error; TBA, 2-thiobarbituric acid.

* Corresponding author. Tel.: +91 11 26059688; fax: +91 11 26059663.

E-mail addresses: sraisuddin@jamiahamdard.ac.in, sheikhraisuddin@yahoo.com (S. Raisuddin).

ity mainly due to their anti-oxidant action (Abd-Allah et al., 2005; Haque et al., 2003; Ishikawa et al., 1989; Sharma et al., 2001; Topal et al., 2005).

There is an overall decrease in the glutathione content in various tissues as a result of CP treatment (Haque et al., 2003; Lopez and Luderer, 2004; Patel and Block, 1985). In urinary bladder also the reduction in the levels of GSH (reduced glutathione) resulting from CP treatment has been reported (Abd-Allah et al., 2005; Manesh and Kuttan, 2005). The reactive metabolites of CP are responsible for various toxic as well as therapeutic actions of CP (Ahmed and Hombal, 1984; Fleming, 1997; Kehrer and Biswal, 2000). Among these metabolites, phosphoramidate mustard is specifically associated with immunosuppressive action of CP (Ahmed and Hombal, 1984; Boddy and Yule, 2000). CP-induced immunosuppression is reported to prompt various types of infection (Angulo et al., 2002; Wijesundera, 1980; Williams and Luzio, 1979). Some of the infectious agents have GSH depleting effects (Hung and Wang, 2004; Rydkina et al., 2004; Suntres et al., 2002). Thus, CP treatment might decrease GSH content itself but the associated secondary infections might cause additional decrease in the GSH level. Therefore, a patient undergoing CP chemotherapy needs excessive supply of GSH restoring anti-oxidants or the compounds, which induce GSH production. A number of GSH-inducing compounds have been found to be effective in reducing CP toxicity in animals (Manesh and Kuttan, 2005).

Plant extracts have also shown protective and/or restorative effect on CP-induced decrease in GSH (Davis and Kuttan, 2000; Haque et al., 2001; Haque et al., 2003; Kumar and Kuttan, 2005; Sharma et al., 2001). In our studies on fenugreek extract (*Trigonella foenum-graecum* L.) we have observed that it is a potent immunostimulating herb in mice (Bin-Hafeez et al., 2003). *T. foenum-graecum* is a popular medicinal herb extensively used as food in Indian sub-continent and has been used in Indian system of medicine for centuries. Its hyperglycemic and hypocholesterolemic effects have also been reported elsewhere (Sharma, 1984; Tayyaba et al., 2001). Chemoprotective effect *T. foenum-graecum* has recently been reported against 7,12-dimethylbenz(alpha)anthracene (DMBA)-induced breast cancer in rats (Amin et al., 2005). Efficacy of fenugreek as a hypoglycemic and anti-diabetic herb has been proven in clinical investigations (Abdel-Barry et al., 2000; Sharma et al., 1990). Therefore, as against a GSH-inducing agent such as *n*-acetyl cysteine or mesna, a herbal extract (fenugreek) with multifaceted action may be an advantageous alternative to purified compounds in preventing CP-induced toxicity.

In the present study we examined restorative effect of *T. foenum-graecum* on CP-induced decrease in GSH. Additionally, assessment of its efficiency was evaluated in reversing depletion of GSH by a known GSH-depleting agent (buthionine sulfoximine, BSO) alone and in combination with CP. The overall aim of this study was to evaluate the efficacy of *T. foenum-graecum* extract in restoration of anti-oxidants and reduction of lipid peroxi-

dation (LPO) in urinary bladder in CP-treated animals which are pre-disposed or committantly exposed to a GSH reducing agent in the form of either infection or use of antibiotics.

2. Materials and methods

2.1. Chemicals

Cyclophosphamide monohydrate [2-[bis-(2-chloroethyl) amino] tetrahydro-2H-1,3,2-oxazaphosphorine 2-oxide monohydrate]; CAS 6055-19-2 and BSO [L-buthionine-SR-sulfoximine]; CAS 5072-26-4, were purchased from Sigma-Aldrich Co., St. Louis, MO, USA.

2.1.1. Plant extract (PE)

Aqueous total extract of seeds of *T. foenum-graecum* L. (Common name – fenugreek; Grade 100, B.No. C/1757) in semisolid form was purchased from the Plant Extract Division of Saiba Industries Pvt. Ltd, Mumbai. The moisture and ash content of extract were 12% and 8%, respectively and pH of 10% aqueous solution of extract was 4.6. Authenticity of extract was certified by the expert taxonomist of the manufacturers. Chemical profile of fenugreek seed extract has been described by various authors and some of the main identified constituents include polyphenols, saponins, alkaloids, fibres, etc. (Dixit et al., 2005; Randhir et al., 2004; Yoshikawa et al., 1997).

2.2. Animals

The study was conducted in male Swiss albino mice (25 ± 2 g) provided by the Central Animal House Facility of the University. The study was approved by the Institutional Animal Ethics Committee (IAEC). The animals were bred and maintained under the standard laboratory conditions (temperature 25 ± 2 °C; photoperiod of 12 h). Commercial pellet diet and water were given *ad libitum*.

2.3. Dosage and experimental groups

BSO, CP and plant extract were suspended in normal saline. Animals were divided in seven groups, Group I–VII. Each group consisted of a minimum of six animals. Control animals (Group I) were administered normal saline p.o. for 10 days and a single i.p. injection of the same on the 10th day. BSO (500 mg/kg body wt.) was administered i.p. on the 10th day (Group II). CP (50 mg/kg body wt.) was given as a single i.p. dose on the 10th day of treatment (Group III). Group IV (BSO + CP) animals were administered BSO i.p. 5 h before CP administration. Group V (CP + PE) animals were administered plant extract for 10 days along with a single i.p. injection of CP on 10th day. Group VI (BSO + PE) animals were given plant extract treatment (100 mg/kg body wt.) p.o. for 10 days and a single i.p. injection of BSO on the 10th day along with the extract. Group VII (BSO + CP + PE) animals were administered plant extract for 10 days and CP and BSO on the 10th day. Dosing was done in such a manner that all the animals could be sacrificed on the same day, i.e., day 11. Selection of doses of BSO and CP are based on pilot experiments using a range of doses and also taking into account previously published report (Bin-Hafeez et al., 2003; Haque et al., 2001, 2003; Ishikawa et al., 1989).

2.4. Biochemical investigations

After termination of treatment, animals were sacrificed under mild anesthesia and their bladders were removed. The bladder tissue was homogenized in chilled phosphate buffer (0.1 M, pH 7.4), using a Potter Elvehjem homogenizer. The homogenate was centrifuged at 10,500g for 30 min at 4 °C to obtain the post-mitochondrial supernatant (PMS), which was used for the biochemical measurements as described below.

2.4.1. Lipid peroxidation

LPO was measured using the procedure of Uchiyama and Mihara (1978). The assay mixture consisted of 0.67% thiobarbituric acid, TBA (HiMedia Labs, Mumbai, India), 10 mM butylated hydroxy toluene, BHT (Amresco Inc., Solon, OH, USA), 1% *ortho*-phosphoric acid (CDH Chemicals, Mumbai, India) and tissue homogenate in a total volume of 3 ml. The rate of LPO was expressed as nmol of TBA reactive substances (TBARS) formed/h/g of tissue using molecular extinction coefficient (ϵ) of $1.56 \times 10^5 \text{ M}^{-1} \text{ cm}^{-1}$.

2.4.2. Measurement of GSH

GSH content was measured in the PMS of urinary bladder by the method of Haque et al. (2003). PMS (1 ml) was precipitated with 1 ml of 4% sulfosalicylic acid (Amresco). The samples were incubated at 4 °C for 1 h and then centrifuged at 1200g for 15 min at 4 °C. The assay mixture consisted of 0.2 ml of filtered aliquot, 2.6 ml of sodium phosphate buffer (0.1 M, pH 7.4) and 0.2 ml 100 mM DTNB (dithio-bis-2-nitrobenzoic acid, Sigma–Aldrich) in a total volume of 3 ml. The absorbance of reaction product was measured at 412 nm and results expressed as nmol GSH/g tissue.

2.4.3. Anti-oxidant enzyme measurements

Glutathione-S-transferase (GST) activity was assayed using the method of Haque et al. (2003). The reaction mixture consisted of 1.675 ml sodium phosphate buffer, 0.2 ml of 1 mM GSH (Sigma–Aldrich), 0.025 ml of 1 mM CDNB (1-chloro-2,4-dinitrobenzene, Amresco) and 0.1 ml of PMS in a total volume of 2 ml. The change in absorbance was recorded at 340 nm and the enzyme activity calculated as nmol CDNB conjugates formed/min/mg protein using ϵ of $9.6 \times 10^3 \text{ M}^{-1} \text{ cm}^{-1}$. GR (glutathione reductase) activity was assayed by the method of Sharma et al. (2001). The assay mixture consisted of 1.6 ml sodium phosphate buffer, 0.1 ml of 1 mM ethylenediamine tetra acetic acid disodium salt (EDTA, Amresco), 0.1 ml NADPH (nicotinamide adenine dinucleotide phosphate reduced, Sigma–Aldrich) and 0.1 ml oxidized glutathione (Sigma–Aldrich) and PMS (0.1 ml) in total volume of 2 ml. The enzyme activity measured at 340 nm was calculated as nmol NADPH oxidized/min/mg of protein, using ϵ of $6.22 \times 10^3 \text{ M}^{-1} \text{ cm}^{-1}$. Glutathione peroxidase (GP) activity was assayed using the method of Sharma et al. (2001). The assay mixture consisted of 1.49 ml sodium phosphate buffer, 0.1 ml EDTA (1 mM), 0.1 ml sodium azide (1 mM, Hi-Media Labs), 0.1 ml of 1 mM GSH (Sigma–Aldrich), 0.1 ml NADPH (0.02 mM), 0.01 ml of 0.25 mM hydrogen peroxide (H_2O_2 , CDH Chemicals) and 0.1 ml PMS in a total volume of 2 ml. Oxidation of NADPH was recorded spectrophotometrically at 340 nm. The enzyme activity was calculated as nmol NADPH oxidized/min/mg of protein, using ϵ of $6.22 \times 10^3 \text{ M}^{-1} \text{ cm}^{-1}$. CAT (catalase) activity was assayed using the method used previously (Haque et al., 2003). The assay mixture consisted of 1.95 ml phosphate buffer, 1 ml H_2O_2 (0.09 M) and 0.05 ml of PMS in a final volume of 3 ml. Change in absorbance was recorded kinetically at 240 nm. CAT activity was calculated in terms of nmol H_2O_2 consumed/min/mg protein.

2.4.4. Protein measurement

Protein was measured by the method of Lowry et al. (1951).

2.5. Statistical analysis

Single factor one-way analysis of variance (ANOVA) was applied to determine significant differences in results of various groups. P values < 0.05 were considered significant. Subsequently, Student–Newman–Keuls test was applied for analyzing the significance between different treatment groups. The values are expressed as mean $\pm$ SE.

3. Results

No mortality and significant change in the body weight were observed in different groups of animals.

3.1. Lipid peroxidation

BSO treatment resulted in a significant increase ($P < 0.01$) in LPO in bladder over control values (Fig. 1). CP treatment also induced LPO significantly. The cumulative effect of BSO + CP resulted in significant increase in the LPO as compared to Group I (control). BSO + CP group (Group IV) showed significant ($P < 0.01$) increase in the levels of LPO when compared to CP (Group III) alone (Fig. 1). Animals pre-treated with PE and subsequently exposed to BSO (BSO + PE) showed a significant ($P < 0.01$) reduction in LPO in bladder. PE treatment also significantly ($P < 0.05$) reduced levels of LPO in bladder when the data of CP + PE group (Group V) was compared with that of CP group (Group II). As shown in Fig. 1, when the data of LPO of BSO + CP group (Group IV) and BSO + CP + PE treatment group were compared, a significant decrease ($P < 0.01$) was observed in case of later (Group VII).

3.2. Reduced glutathione

Fig. 2 shows cellular GSH data of urinary bladder. A significant ($P < 0.01$) decrease in GSH was observed in BSO, CP and BSO + CP-treated groups (1.3, 1.8 and 0.8 nmol GSH/gm tissue, respectively) as compared to control values (2.5 nmol GSH/gm tissue). Also, there was a significant ($P < 0.01$) decrease in the levels of GSH in BSO + CP group (Group IV) when compared to CP group (Group III). As shown in Fig. 2, when comparison was made between BSO and BSO + PE groups, GSH content was found to be increased significantly ($P < 0.01$) in case of later. Similarly, bladder of CP + PE group (Group V) showed significantly increased ($P < 0.01$) GSH content as compared to CP alone group (Group III). When the data of GSH of BSO + CP (Group IV) group was compared with BSO + CP + PE group (Group VII), significant ($P < 0.01$) restoration in GSH was recorded (Fig. 2).

3.3. Anti-oxidant enzymes

BSO and CP treatments significantly ($P < 0.01$) decreased the activities of GST, GR, GP and CAT in the bladder when compared with control group (Table 1). BSO + CP group also showed a significant ($P < 0.01$) decrease in the activities of GST, GR and GP when compared with CP or control group. However, there was no significant difference in CAT activity between the animals of group IV (BSO + CP) and group III (CP). Activities of these anti-oxidant enzymes increased significantly ($P < 0.01$) in both BSO + PE (Group VI) and CP + PE (Group V) groups when compared to their respective controls, viz. BSO (Group II) and CP (Group III) (Table 1). Animals treated with PE and subsequently exposed to BSO + CP (Group VII, BSO + CP + PE treatment) displayed a significant increase ($P < 0.01$) in the activities of

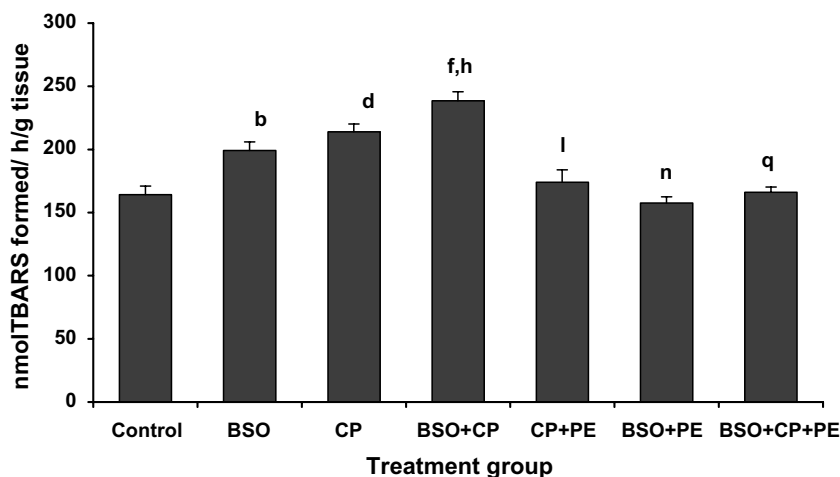


Fig. 1. Effect of *T. foenum-graecum*, BSO and CP on the lipid peroxidation in urinary bladder of mice. Significant differences are indicated by ^b $P < 0.01$ and ^d $P < 0.01$ in Group II (BSO) and Group III (CP) treated animals, respectively and ^f $P < 0.01$ in Group IV (BSO + CP), when compared with control animals (Group I). ^b $P < 0.01$ indicates significant levels of BSO + CP group when compared to CP group. ^l $P < 0.01$ and ⁿ $P < 0.05$ indicate significant difference of data of Group V (CP + PE) and Group VI (BSO + PE) when compared with Group III and Group II, respectively. ^q $P < 0.01$ when Group IV data was compared with Group VII. Values are means $\pm$ SE ($n = 6$).

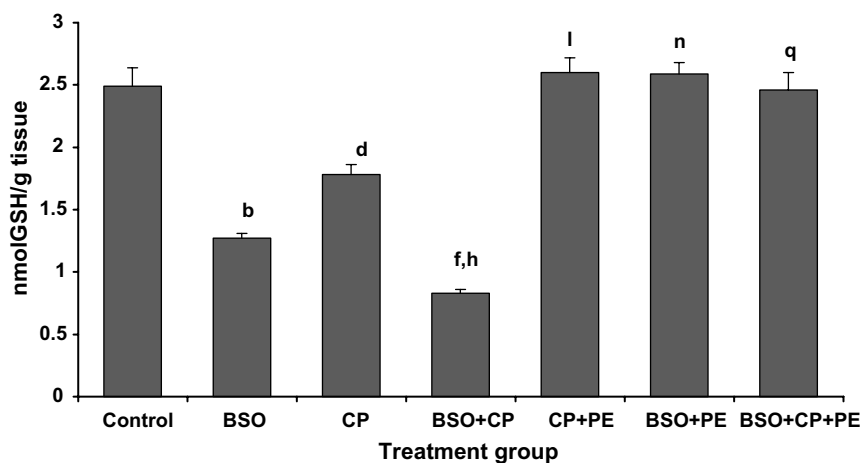


Fig. 2. Effect of *T. foenum-graecum*, BSO and CP on GSH content in urinary bladder of mice. Significant differences are indicated by ^b $P < 0.01$ and ^d $P < 0.01$ in Group II (BSO) and Group III (CP) animals, respectively and ^f $P < 0.01$ in Group IV (BSO + CP) when compared with control animals (Group I). ^b $P < 0.01$ indicates significant change in BSO+CP group when compared with CP group. ^l $P < 0.01$ and ⁿ $P < 0.01$ indicate significant differences in observations in Group IV (CP + PE) and Group VI (BSO + PE) when compared with GSH data of Groups III and II, respectively. ^q $P < 0.01$ when GSH data of Group IV was compared with Group VII. Values are means $\pm$ SE ($n = 6$).

Table 1

Activities of anti-oxidant enzymes in the urinary bladder of mice in different treatment groups

Group	Activity of anti-oxidant enzyme			
	GST	GR	GP	CAT
I (Control)	141 $\pm$ 3	141 $\pm$ 4	153 $\pm$ 5	102 $\pm$ 3
II (BSO)	97 $\pm$ 2 ^b	90 $\pm$ 4 ^b	120 $\pm$ 3 ^b	95 $\pm$ 2 ^b
III (CP)	112 $\pm$ 3 ^d	120 $\pm$ 5 ^d	137 $\pm$ 4 ^c	82 $\pm$ 3 ^d
IV (BSO + CP)	76 $\pm$ 3 ^{f,h}	84 $\pm$ 2 ^{f,h}	116 $\pm$ 3 ^{f,h}	80 $\pm$ 3 ^f
V (CP + PE)	150 $\pm$ 5 ^l	150 $\pm$ 5 ^l	160 $\pm$ 2 ^l	105 $\pm$ 5 ^l
VI (BSO + PE)	144 $\pm$ 3 ⁿ	147 $\pm$ 6 ⁿ	154 $\pm$ 4 ⁿ	111 $\pm$ 7 ⁿ
VII (BSO + CP + PE)	142 $\pm$ 4 ^q	146 $\pm$ 6 ^q	152 $\pm$ 4 ^q	102 $\pm$ 5 ^q

Values are means $\pm$ SE ($n = 6$). GST expressed as nmol CDNB conjugates/min/mg protein, GR as nmol NADPH oxidized/min/mg of protein, and GP as nmol NADPH oxidized/min/mg protein. CAT activity is expressed as nmol H₂O₂ consumed/min/mg protein. Significant differences are indicated by ^b $P < 0.01$, ^d $P < 0.01$, ^c $P < 0.05$ and ^f $P < 0.01$ when compared with control animals. (Group I) ⁿ $P < 0.01$ when compared with Group II, ^h $P < 0.01$ and ^l $P < 0.01$ when compared with CP-treated animals (Group III) and ^q $P < 0.01$ when compared with Group IV.

all the anti-oxidant enzymes when compared with BSO + CP group (Group IV).

4. Discussion

Urotoxicity of CP is a well-established side effect (Cox, 1979; Davis and Kuttan, 2000; Manesh and Kuttan, 2005). Since bladder is the site for the storage of urine, concentration of toxic metabolites of CP is higher in bladder as compared to other organs, which increases the toxic load of CP in bladder to manifold (Beyer-Boon et al., 1978). CP caused a significant reduction in the levels of all the anti-oxidants in urinary bladder. Besides, it increased LPO in the bladder. The oxidative products of the CP responsible for induction of LPO and generation of the ROS result in inflammation thus disturbing the overall redox cycling of the bladder (Cooper et al., 1986; Kehrer and Biswal, 2000). BSO, a known depletor of glutathione, inhibits γ -glutamylcysteine synthetase (γ -GCS) (Griffith, 1982). When data of reduction of GSH were compared, it was found that BSO has more significant effect than CP. A difference of 42% was noticed between the two observations confirming that BSO is a more potent depletor of GSH than CP as reported elsewhere (Ishikawa et al., 1989). CP-induced depletion of GSH is primarily mediated by interaction of its reactive metabolite, acrolein with GSH (Kehrer and Biswal, 2000). Acrolein not only interacts with GSH but also with cysteine which is one of the constituent amino acids of GSH (Kehrer and Biswal, 2000). Therefore, a number of sulfhydryl (–SH) compounds and cysteine itself have been observed to protect the animals from toxic effects of CP (Ishikawa et al., 1989; Manesh and Kuttan, 2005; Roberts et al., 1994).

Intraperitoneal administration of CP-induced LPO in bladder in a significant manner. CP-induced LPO has been reported in different tissues of exposed animals (Haque et al., 2003; Patel, 1987; Selvakumar et al., 2005; Venkatesan and Chandrakasan, 1995). In CP-induced LPO also role of acrolein has been implied (Adams and Klaidman, 1993; Uchida, 1999). It has been suggested that by binding to nucleophilic amino acids, acrolein could directly affect transcription as well as modulate this process through its ability to deplete GSH (Kehrer and Biswal, 2000). BSO treatment resulted in depletion of GSH and increase in LPO in urinary bladder. Depletion of GSH is also reported to increase susceptibility of cells to apoptosis (Mirkovic et al., 1997; Zucker et al., 1997).

When BSO and CP were administered together, an additive effect was observed in case of GSH, LPO and other parameters. The purpose of using BSO along with CP was to study a likely scenario where host is exposed to a combination of GSH depleting agents including pathogens and to assess whether herbal extract treatment of fenugreek has any modulatory effect on their commutative/additive effect. Fenugreek extract pre-treatment not only showed protective effect on CP urotoxicity but was also effective in protecting the animals treated with CP + BSO combina-

tion (Group VII). While herbal extract pre-treatment restored depleted GSH and other anti-oxidants, simultaneously it reduced LPO in the bladder. CP-induced immunosuppression is likely to increase incidence of infections which may deplete GSH as many infectious agents are reported to deplete GSH (Hung and Wang, 2004; Rydkiina et al., 2004; Suntres et al., 2002). An immunomodulatory herbal extract like fenugreek which has GSH restoring effect holds great promise in reducing the adverse effects of CP in cancer patients.

T. foenum-graecum extract has demonstrated immunomodulatory effect in mice (Bin-Hafeez et al., 2003). A number of herbal extracts and their isolated constituents have also shown protective effect against CP-induced urotoxicity (Davis and Kuttan, 2000; Nair et al., 1993; Vieira et al., 2004). Thiols containing compounds such as mesna and cysteine have shown protective effects against CP-induced urotoxicity (Roberts et al., 1994). Fenugreek extract has been extensively used in a number of herbal formulations. For centuries, it is being used as a folklore medicine. Its various pharmacological actions reported hitherto in ancient medicine literature have been scientifically validated later. These include its anti-inflammatory, antipyretic, hypoglycemic and immunomodulatory activities (Bin-Hafeez et al., 2003; Sharma, 1984).

Flavonoids of fenugreek extract have been observed to possess anti-oxidant activity (Moskaug et al., 2005; Myhrstad et al., 2002; Ozcan et al., 2005). For example, quercetin has shown protective effect against CP-induced hemorrhagic cystitis (Ozcan et al., 2005). Moreover, in a recent study fenugreek seed extract has been reported to prevent both LPO and hemolysis in RBC (Kaviarasan et al., 2004). Fenugreek seeds have also been reported to raise the anti-oxidant levels and lower the LPO in liver of ethanol intoxicated (Thirunavukkarasu et al., 2003) and diabetic rats (Anuradha and Ravikumar, 2001). The findings of the present investigation demonstrate that fenugreek extract pre-treatment prevented CP urotoxicity which is primarily mediated by LPO and depleted GSH by reversing these effects. Moreover, because of its widespread availability and relative lack of toxicity, fenugreek deserves consideration for use in cancer patients as a complementary therapy after further validation and trials. This observation is strengthened by a number of clinical trials showing the efficacy of fenugreek as a hypoglycemic and anti-diabetic agent with limited toxic manifestations (Bordia et al., 1997; Gupta et al., 2001; Sharma et al., 1990).

Conflict of interest statement

There is no conflict of interest with any person or other organization.

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